Pranav Kulkarni

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SUMMARY

Computer Science and Engineering undergraduate at B.M.S. College of Engineering (Diploma in CSE, CGPA 9.55/10) who builds full-stack and AI applications in Python, Java, React and TypeScript, with a focus on secure, well-tested backends. Seeking Software Engineering Internship roles.

EDUCATION

B.M.S. College of Engineering, Bengaluru — Sep 2026 – Expected 2029
Bachelor of Engineering in Computer Science and Engineering

Vasantrao Potdar Polytechnic, Belagavi — Jul 2023 – Jun 2026
Diploma in Computer Science and Engineering — CGPA: 9.55/10 (A+)

Coursework (ongoing): Data Structures in C (github.com/pk8481/DS), Object-Oriented Programming

TECHNICAL SKILLS

Languages: Python, Java, C, TypeScript, JavaScript, HTML, CSS

Frameworks and Libraries: React, Vite, Tailwind CSS, FastAPI, Pydantic, Streamlit, PyTorch, TensorFlow, scikit-learn, XGBoost, Pandas, OpenCV

AI and Machine Learning: Machine Learning, Computer Vision, Speech-to-Text, Voice Biometrics, Large Language Models (LLMs)

Security: Biometric Authentication, JWT, OAuth 2.0, Encryption at Rest (Fernet), Role-Based Access Control, Rate Limiting, Audit Logging

Tools and Practices: REST APIs, SQLite, Git, GitHub Actions (CI), Docker, Unit Testing

PROJECTS

BioAuth – Face and Voice Biometric Passwordless Authentication | Project Walkthrough — Dec 2025 – Feb 2026
React, TypeScript, Python, FastAPI, PyTorch, InsightFace, SpeechBrain, faster-whisper, SQLite, Docker

- Built a full-stack multi-factor passwordless login combining a face match (InsightFace), speaker verification (SpeechBrain ECAPA) and a spoken secret phrase transcribed with faster-whisper and fuzzy-matched.
- Designed a FastAPI backend with 49 REST endpoints, JWT sessions, three-tier role-based access, per-client rate limiting and progressive lockouts (10 min, 30 min, 2 h) against brute-force attacks.
- Protected biometric data with Fernet encryption at rest, stored secret phrases only as hashes and HMAC tokens, and added consent-based enrollment, automatic data retention cleanup and a full audit trail.
- Wrote 60+ automated backend tests run in a GitHub Actions CI pipeline; models run on CUDA GPUs with CPU fallback, and the app is containerized with Docker for cloud deployment.
- Integrated Google OAuth sign-in and built an admin console and a Gemini-powered security assistant with automatic fallback across three AI providers.

House Price Prediction and Real Estate Intelligence System | AIML skill lab (HCL) — Oct 2026
Python, scikit-learn, XGBoost, TensorFlow, Pandas, FastAPI, Streamlit, Docker, pytest

- Cleaned 13,320 Bengaluru property listings (unit conversion, duplicate and leakage removal, validity rules) and compared 7 models with 5-fold cross-validation; a tuned Gradient Boosting model reached 15.6% median error on a held-out test set, against 39.9% for a baseline.
- Served predictions through a FastAPI service and a Streamlit app with price ranges (81% test coverage) and GenAI explanations that reject invented numbers; built test-first with 143 pytest tests and containerized with Docker.

PawCare – Stray Animal Rescue and Adoption Platform (Team project, 4 members) — Oct 2026 – Present
React, Node.js, Express, MongoDB, JWT, Leaflet

- Team-built platform connecting people who report stray or injured animals with volunteers, rescuers, shelters and adopters, with role-based workflows for each user group.
- Owned testing and quality assurance: tested the app end to end and found 4 key issues: a security gap where accounts were never verified, missing role selection at sign-up, uploaded animal photos not displaying, and a hard-to-use map location picker.
- Reported issues to the team and contributed to the frontend design.

Facial Emotion Recognition – Real-Time Face Analysis — *Python, OpenCV, TensorFlow/Keras, DeepFace, dlib*

- Trained a Keras CNN that classifies 7 facial emotions and built a real-time webcam system around it, compared live against DeepFace, with age and gender estimation and face login protected by blink-based (eye aspect ratio) liveness detection.

Exam Seating Arrangement System – automates exam scheduling and seating allocation with custom allocation logic.

CERTIFICATIONS

Artificial Intelligence and **Cyber Security Foundation** – Infosys Springboard (Aug 2026) · **Digital Productivity with AI** – UNICEF YuWaah (Aug 2026, score 100%) · View certificates